CS105 Midsem: DIC on Discrete Structures

45 marks, 120min

21 Sep 2024

Instructions:

- Attempt *all* questions. Write all answers and proofs carefully. You can assume results shown in class. However, if you are *making any such assumptions or using results proved in class, state them clearly.*
- All sets considered below are general sets that can be infinite. Hence, you must not assume that they are finite or countable, unless clearly specified otherwise.
- Recall that $\mathbb{R}, \mathbb{Q}, \mathbb{Z}, \mathbb{N}$ denote, respectively, the set of real numbers, rational numbers, integers and natural numbers.
- In this course, one of our aims is to learn how to write good proofs, hence considerable weightage will be given to *clarity and completeness* of proofs.
- Do *not* copy or use any other unfair means. Offenders will be reported to the Disciplinary Action Committee.

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1. (2+3=5 marks) Let X and Y be finite sets with $|X| = 10, |Y| = 2$. Then
   (a) What is the number of functions from X to Y?
   (b) What is the number of surjections from X to Y?

   Give a short justification for your answers.

2. (2+3+4=9 marks) Prove or disprove:
   (a) There exists a natural number $n \geq 1$ such that $n^2 + 3n + 2$ is prime.
   (b) For any positive integer $n \geq 2$ if there is no prime number p, such that $1 \leq p \leq \sqrt{n}$ which divides n, then n must be prime.
   (c) For every positive integer n, there exists a consecutive sequence of n positive integers which are all composite.

3. (2.5+2.5=5 marks) Write True or False. If true, provide a proof. Else give a counter-example.
   (a) If X is a finite set and $f : X \to X$ is an injection, then f is a bijection.
   (b) If X is a countably infinite set and $f : X \to X$ is an injection, then f is a bijection.

4. (4 marks) What is the number of equivalence relations on a set of 4 elements? Explain your reasoning.

5. (3+2+5=10 marks) Let $f : A \to B$ be a function. Let us define the relation R_f by xR_fy iff $f(x) = f(y)$.

(a) Prove that R_f is an equivalence relation.

(b) Do there exist two distinct functions $f, g : \mathbb{R} \to \mathbb{R}$ such that $R_f = R_g$? If yes provide these functions with a justification, else prove that it is impossible to have such functions.

(c) The fiber of the function f, is defined by $fib(f) = \{f^{-1}(b) \mid b \in B\} = \{\{a \in A \mid f(a) = b\} \mid b \in B\}$. Let f be a surjective function. Then what is the relation between equivalence classes of R_f and $fib(f)$? Does this hold for any arbitrary function f? Why or why not?

6. (5+5+2=12 marks) Let $S = \{I_1, \ldots I_n\}$ be a set of n open intervals on the real line with rational endpoints. That is, for all $1 \leq j \leq n$, $I_j = (a_j, b_j)$ for some $a_j, b_j \in \mathbb{Q}$.

(a) Suppose we know that any two of the intervals have a non-empty intersection, i.e., $I_i \cap I_j \neq \emptyset$ for all $1 \leq i, j \leq n$. Prove by induction that the intersection of all intervals in S is non-empty, i.e., $\bigcap_{j=1}^{n} I_j \neq \emptyset$.

(b) Suppose instead that we only know that for some positive integer k, among any $k + 1$ of the intervals from S, there are two with a non-empty intersection. Then, prove that there exists a set of k points on the real line such that each interval in S contains at least one of them.

(c) Suppose set S is a countably infinite set of open intervals. Then does the statement in (b) above hold? If yes, prove it, else give a counter-example.